

CSF340

Assignment-1

ID:- 23201003 Section: 23

Masyar Sarar

1)

Redundancy improves system reliability, availability, and fault tolerance, by providing backup components or data that can take over if primary ones fail.

We use tractor-trailer, since the dual tires on each side of its rear axle allow the truck to continue driving even when one tire fails.

2)

a) Geometric mean gives same ~~result~~ relative result no matter what computer is used. ~~to~~ If we use arithmetic mean then the result would vary depending on computer. It would be bias towards the large value.

b) SPEC ratio greater than 1 means the ~~the~~ tested system is greater than ~~the~~ the reference system.

$$c) \text{SPEC ratio} = \sqrt[4]{\frac{250}{200} \times \frac{180}{120} \times \frac{300}{400} \times \frac{90}{60}} = 1.205$$

The SPEC ratio 1.205 means The CPU-2 is 1.205 times faster than the reference system.

3) a)

$$\text{Alpha CPU time} = 1.5 \times 10^9 \times 3.2 \times 2 \times 10^{-9} = 9.6 \text{ s}$$

$$\text{Beta CPU time} = 1.5 \times 10^9 \times 2.8 \times 2.5 \times 10^{-9} = 10.5$$

b) Alpha's CPU performance is better.

$$\text{Alpha is } \frac{10.5}{9.6} = 1.094 \text{ times faster.}$$

c) I would recommend Alpha CPU Model because it has lower CPU time

d) a)

4) Amdahl's law: $\equiv$ The performance enhancement is possible with a given improvement is limited by the amount that the improved feature is used.

b)

Yes, Amdahl's law is connected to design - 3:
"Make common case faster"

If ~~I~~ you optimize the task that occurs most frequently then overall performance will be more increased rather than if we optimize rare case.

Example:-

If 95% of execution time is spent in a particular loop then optimizing that loop will yield a much higher gain than optimizing other 5%.

$$c) T_{\text{improved}} = \frac{10 \times 40\%}{4} + 10 \times 60\% = \frac{4}{4} + 6 = 7s$$

Initial time = 10s

Improved time = 7s

So compared to the initial time, the improved time is $\frac{10}{7} = 1.42$ times faster.

b)

a) For X-CPU

$$\text{Average CPI} = 0.4 \times 3 + 0.4 \times 4 + 0.2 \times 2 = 3.2$$

For Y-CPU

$$\text{Average CPI} = 0.4 \times 2 + 0.4 \times 3 + 0.4 \times 2 = 2.4$$

b)

$$\text{Clock Cycle} = \text{CPI} \times \text{Instruction Count}$$

$$\text{X-CPU's Clock cycle} = 5 \times 10^5 \times 3.2 = 1.6 \times 10^6 \text{ cycle}$$

$$\text{Y-CPU's clock cycle} = 5 \times 10^5 \times 2.4 = 1.2 \times 10^6 \text{ cycle}$$